Needs-based recommendation systems

Machine Learning Pipeline for Financial Needs Prediction and Product Recommendation

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3. ML Workflow & Hyperparameter Optimization & Explainable AI (XAI)
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Project Flow

1. Exploratory Data Analysis
Data quality, distributions, outliers

↓

2. Classification & Model Selection
Baselines, tuning, explainability, DCN comparison

↓

3. Recommendation Engine
SVD/Autoencoder ranking and product matching

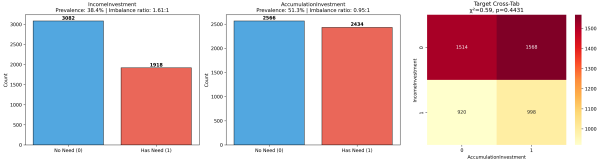
Target Variable Analysis

Key Findings

- `AccumulationInvestment`: **balanced** (51.3%, $\approx 1{:}1$)
- `IncomeInvestment`: **moderate imbalance** (38.4%, $\approx 2.3{:}1$)
- χ^2 test: $p = 0.44 \gg 0.05 \Rightarrow$ targets are independent.

Modeling Implication

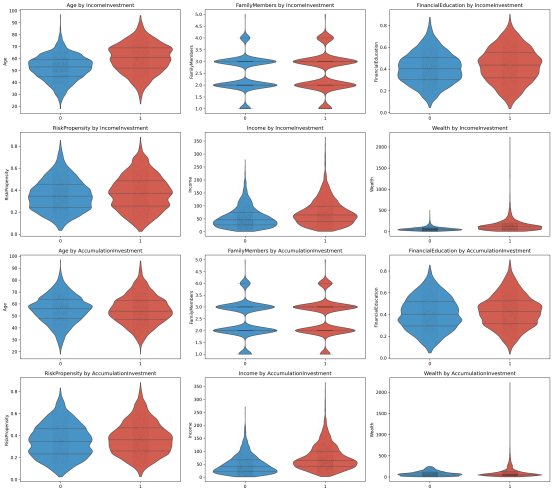
Naive all-zero baseline gives $\sim 70\%$ accuracy, so accuracy alone is misleading. We optimize $\mathbf{F}_{0.5}$ and model the two targets as outputs.



Target prevalence and cross-tab evidence

Feature Distributions by Target

- `Age`, `Income`, `Wealth`, `RiskPropensity`: distributions are **significantly higher for class 1**, indicating that these variables strongly discriminate between the two groups
- `FinancialEducation`: **moderate upward shift** for class 1, suggesting a secondary but positive effect
- `FamilyMembers`: **no visible difference**



Outlier Detection — Isolation Forest

Method choice

- **Isolation Forest** (300 estimators, `random_state=42`)
- Features standardised (StandardScaler) before fitting
- `Gender` (binary): inclusion does not distort the geometry
- Threshold: **5th percentile** of anomaly scores
  $\Rightarrow$ flags $\approx 5\%$ of observations as outliers

Decision: outliers retained

- Extreme `Income` / `Wealth` values are **valid** business profiles, not data errors
- Preferred treatment: **robust scaling** +

2D projections (Isolation Forest)

Key observations from scatter plots

Income vs Wealth: outliers concentrated at very high values — long right tail confirms extreme inequality.

Age vs Income / Wealth: older clients more likely to be flagged — plausible, not errors.

t-SNE projection: outliers (red) form a sparse fringe around the main cluster (blue). No clear isolated sub-group.

Conclusion: high-value financial profiles are structurally different, not corrupted data.

Feature Transformation and Engineering

1. Data Transformation (`DataTransformer`)

- **Numeric Conversion:** Coercion of mixed/string formats to continuous variables.
- **Log Transformation:** Applied $\log(1+x)$ to highly skewed financial variables (`Wealth`, `Income`) to normalize distributions.
- **Scaling:** `StandardScaler` applied post-engineering for stable convergence.

2. Household Metrics

- `WealthIncomeRatio`
- `IncomePerFamilyMember`

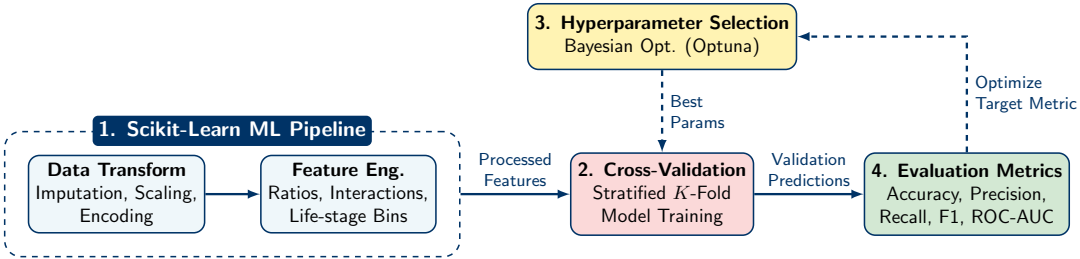
3. Interaction Features

- `RiskEducationInteraction` ($Risk \times Edu$)
- `RiskWealthInteraction` ($Risk \times \log(Wealth)$)
- `AgeRiskInteraction` ($Age \times Risk$)

4. Non-linearity & Life-Stages

- `AgeSquared`: Models the "Life-Cycle" curve (wealth peaking in mid-life).
- **Age Bins**: One-hot flags for `Under35`, `35_54`, `55_69`, and `70plus`.

ML Pipeline Overview



Baseline Models Evaluated

Models Evaluated

We comprehensively tested several families of classifiers across both target variables (*Income* and *Accumulation*):

- **Distance/Linear**: SVM, Naive Bayes, KNN.
- **Tree-based Ensembles**: Random Forest, XGBoost, LightGBM.
- **Meta-Ensembles**:
  - Voting Classifier (Soft & Hard)
  - Stacking Classifier (LGBM, XGB, KNN, RF with a Logistic Regression meta-learner)

Top Performers & Selection

Selected for Tuning:

- **Random Forest**
- **LightGBM**

Selection Rationale:

- Strong baseline predictive power with robust handling of non-linear interactions.
- **Target Metric:** Optimised for **F0.5 Score** to prioritise precision over recall (minimising the risk of unsuitable, false-positive product recommendations).

Optuna Hyperparameter Tuning

Objective Metric

- Optimized for **F0.5 Score** (Stratified 5-Fold CV)
- *Rationale:* Prioritizes precision over recall ($\beta = 0.5$), significantly penalizing false positives (bad recommendations).

TPE Sampler Strategy

- Uses Tree-structured Parzen Estimator (TPE).
- Explores the search space efficiently by modelling the probability of trial performance.

Hyperparameter Search Space

Random Forest (20 Trials)

- `n_estimators`: 10–200
- `max_depth`: 3–10
- `min_samples_split/leaf`: Constraints for regularization.

LightGBM (50 Trials)

- `learning_rate`: 0.001–0.2 (Log scale)
- `num_leaves`: 15–255
- `reg_alpha` & `reg_lambda`: L1/L2 Regularization (Log scale)

Optimization Results

- The tuned models demonstrated a significant reduction in false positives while maintaining competitive accuracy.

	Income Target			Accumulation Target		
Model	**Precision**	**Recall**	**F1 Score**	**Precision**	**Recall**	**F1 Score**
Baseline RF	0.7953	0.5260	0.6332	0.8301	0.7622	0.7947
Tuned RF	**0.8361**	0.5312	0.6497	**0.8581**	0.7544	0.8029
Baseline LGBM	0.8402	0.5339	**0.6529**	0.8503	0.7641	0.8049
Tuned LGBM	**0.8909**	0.5104	0.6490	**0.8568**	**0.7700**	**0.8111**

Final Selection

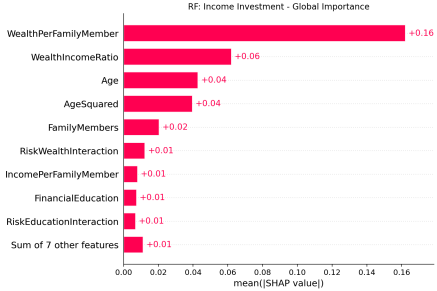
For the final recommendation pipeline, we use **Tuned RF** for **Income Investment** and **Tuned LightGBM** for **Accumulation Investment**, balancing precision and task-specific predictive strength.

XAI Methods: SHAP, LIME & Permutation

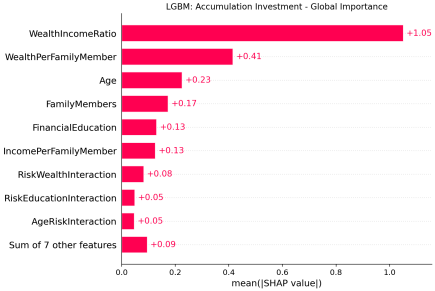
- **SHAP (SHapley Additive exPlanations):** Used to understand global feature importance and directionality across the entire dataset. It highlights which engineered features drive predictions the most.
- **LIME (Local Interpretable Model-agnostic Explanations):** Used to explain individual, client-level predictions to build trust and provide clear reasoning for a recommended product.

LightGBM Model Insights: Income vs Accumulation

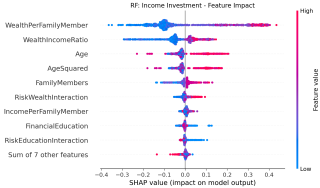
Income Investment



Accumulation Investment



SHAP Feature Importance



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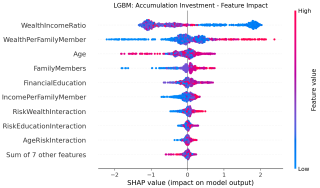
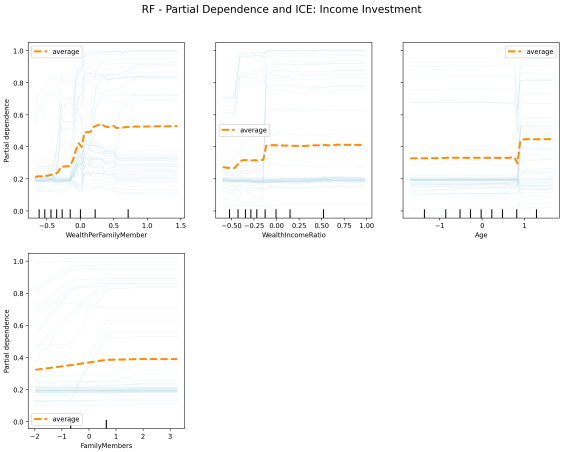
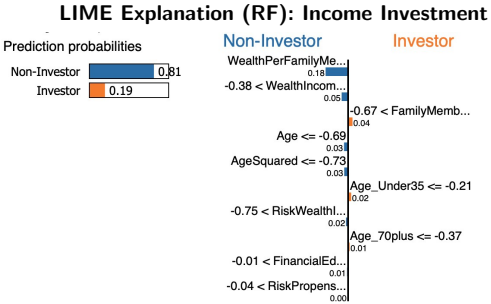


SHAP Beeswarm Plot

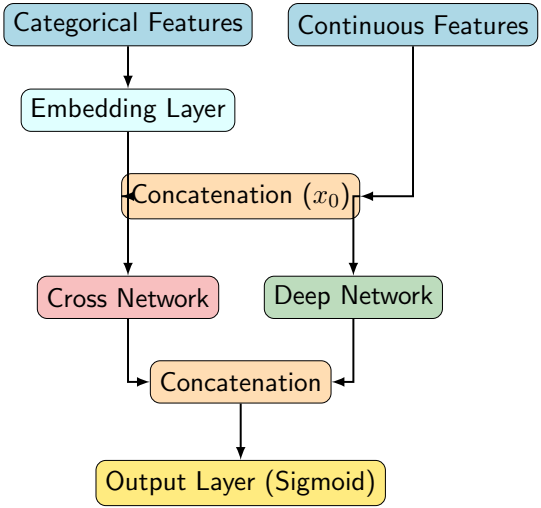
SHAP Beeswarm Plot

LIME, PDP and ICE for RF income investment





DCN Overview and Selection Decision [1]



DCN Results

Task	Acc	Prec	Rec	F1	AUC
Income	0.776	0.827	0.547	0.659	0.765
Accum.	0.795	0.851	0.702	0.769	0.848

Why Not Selected

- Slight gain for **Income** vs tuned RF for F1, but not for precision.
- Lower performance for **Accumulation** vs tuned LightGBM.
- Lower interpretability than tree-based models with SHAP/LIME.

Methodology: User-Item Matrix

We construct a user-item interaction matrix R using predicted probabilities from the supervised classifiers, penalised by risk misalignment.

Matrix Entry R_{ij}

$$R_{ij} = P_{\text{type}(j)}(i) \cdot \big(1 - \alpha \cdot |\text{RiskPropensity}(i) - \text{Risk}(j)|\big)$$

- $P_{\text{type}(j)}(i)$: Predicted probability of client i needing product j's type (Accumulation or Income).
- $\alpha \in [0, 1]$: Decay factor penalising risk misalignment.
  - **Economic Rationale**: Reflects the firm's risk aversion and regulatory suitability constraints (e.g., MiFID II).
  - **Trade-off**: Balances maximising expected fee revenue/conversion against minimising compliance risks and long-term customer churn from unsuitable investments.

Methodology: Models & Evaluation

We use model-based CF [2] leveraging matrix R:

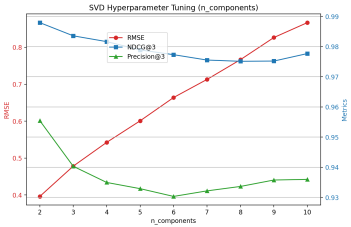
- **Truncated SVD**: Factorises $R \approx U_k \Sigma_k V_k^T$ into low-dimensional latent factors to capture hidden preferences.
- **Autoencoder**: Learns non-linear representations by compressing and reconstructing the interaction matrix ($z_u = f_\theta(r_u)$, $\hat{r}_u = g_\phi(z_u)$).

Evaluation Metrics

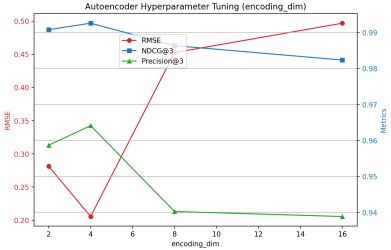
- **RMSE**: Measures the model's reconstruction error.
- **Precision@k & Recall@k**: Evaluate the relevance of the top-k recommendations.
- **NDCG@k**: Assesses ranking quality, prioritising relevant items placed higher.

Experimental Results

Model	RMSE	Precision@3	Recall@3	NDCG@3
Truncated SVD	0.396	0.955	0.955	0.988
Autoencoder	**0.206**	**0.964**	**0.964**	**0.993**



SVD Tuning



Autoencoder Tuning

Financial Perspective

The Autoencoder's superior ranking quality (higher Precision@3/Recall@3/NDCG@3) supports better product-client matching, which can improve conversion and fee revenue while reducing suitability risk and long-term churn from mis-sold recommendations.

References

[1] Weiyu Guo et al. "Explainable enterprise credit rating using deep feature crossing". en. In: *Expert Systems with Applications* 220 (June 2023), p. 119704.

[2] Marwa Sharaf et al. "A survey on recommendation systems for financial services". In: *Multimedia Tools and Applications* 81.12 (May 2022), pp. 16761–16781.